

Exam Date & Time: 06-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Computer Vision [ECE-4051]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compute the output image I_{out} obtained by applying 3 x 3 uniform smoothing operator on the 8-bit grayscale image I_{in} shown in Figure 1. Using the output image I_{out} as an example, identify which operator (Uniform or Gaussian) is a better approximation of a defocused camera. Justify your answer.

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	11	11	0	0	0
0	0	0	11	11	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

(5)

Figure 1. The input image I_{in}

- B) Discuss the effect of varying sigma of Gaussian averaging filter to reduce noise in an image. (3)
- C) Draw the histogram of the given 8-bit image and comment on the nature of the image. Justify your reasoning. (2)

$$Img = \begin{bmatrix} 32 & 32 & 48 & 48 & 64 \\ 0 & 16 & 0 & 16 & 48 \\ 32 & 0 & 32 & 32 & 48 \\ 32 & 16 & 16 & 16 & 64 \\ 0 & 0 & 32 & 48 & 64 \end{bmatrix}$$

(2)

- 2) Discuss how the Scale-Invariant Feature Transform (SIFT) descriptor algorithm extract local features from an image. Explain whether the SIFT detector and descriptor be used together. Justify your answer. (5)

A)

B)

(3)

Construct GLCM $C_{(1,2)}$ and $C_{(2,2)}$ for the given image.
Discuss what information is obtained from these two matrices?

$$Img = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 1 \\ 1 & 1 & 2 & 2 \end{bmatrix}$$

- C) Describe the method of local texture representation using filters. (2)
- 3) Design a segmentation algorithm to segment greenery (grass, trees, etc) in the following image. Clearly describe each step of the proposed algorithm. Discuss the possible limitations of the algorithm and the possible areas/regions which might not be accurately segmented. Justify your answer.
- A)



- (5)
- B) With the help of a suitable diagram, explain epipolar constraint. (3)
- C) Define Essential Matrix and explain its usefulness in binocular reconstruction. (2)
- 4) Given the density estimate

A)

$$P(x) = \frac{1}{n} \sum_{i=1}^n K(x - x_i)$$

for finite set of data points $x_1, x_2, \dots, x_n$ and $K(x)$ is the kernel. Show that the mean shift vector has the direction of the gradient of the density estimate. Describe in detail each step of mean shift clustering algorithm.

B)

(3)

Given the observed data $(x_i, y_i), i = 1, 2, \dots, N$, we wish to fit a line using probabilistic models. Assume the following model: x coordinate is generated from a uniform distribution and y coordinate is generated by finding the point $ax_i + b$ on the line corresponding to the x coordinate then adding a zero mean normally distributed random variable. Show that maximizing the likelihood of the data is equivalent to least square line fitting.

- C) With the help of a suitable example, discuss the effect of noise in fitting line using Hough Transform. (2)
- 5) Compare kNN classifier with linear SVM for image classification, with the help of suitable diagrams. (5)
- A)
- B) A classifier was trained on a dataset of cats (Class 1) and dogs (Class 0) and was tested on 750 test images, half of which belongs to cats and the other half are images of dogs. The trained classifier correctly classified 300 images of cats and 350 images of dogs. Calculate precision, recall and the confusion matrix of the test results (3)
- C) Explain over-fitting in classifier. Discuss ways in which the over-fitting problem can be reduced. (2)

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